Sustainable Pathways for Resource Management: Comparative Analysis of Lithium-Ion Battery Recycling Methodologies

[redacted]

Abstract: The rapid proliferation of lithium-ion batteries(LIBs) has led to resource scarcity and environmental concerns, prompting significant global attention to recycle spent LIBs. This report introduces the current industrial approaches, evaluating the primary methods in metal recovery: Pyrometallurgy, while scalable and simple, suffers from excessive energy consumption, lithium loss, and toxic emissions. Hydrometallurgy offers low-energy processing, but faces challenges in metal recover rate and reagent management. A promising hybrid "mild recycling" method combines the conventional processes, achieving comprehensive metal recovery with reduced energy and chemical inputs. This analysis highlights the imperative to balance technical feasibility, economic viability, and environmental sustainability in LIBs recycling systems.

Keywords: Lithium-ion battery; Industrial recycling; Resource sustainability

1 Introduction

The exponential expansion of lithium-ion batteries (LIBs) in contemporary society represents a technological triumph of worldwide energy transition, while precipitating environmental and resource challenges. The global demand for LIBs is projected to increase tenfold between 2020 and 2030, primarily driven by the prospering market for electric vehicles and renewable energy storage systems.[1] In 2023, the number of newly registered electric vehicles has increased tremendously by 14 million, prompting worldwide gigafactories to expand their capacity of LIBs.[2] This growth trajectory has magnified concerns regarding the sustainable supply of critical battery materials, including lithium, cobalt, nickel, and manganese.[3] Projections indicate that due to increasing consumption and mining difficulties, cobalt reserves will confront the risk of depletion in 60 years, and lithium provision can be affected within 30 years.[4,5]

The improper management of spent LIBs poses significant environmental threats, leading to soil and aquatic system contamination through heavy metal leaching.[6] Furthermore, recycling end-of-life lithium-ion batteries can effectively mitigate the depletion trend of critical metal resources, preventing supply constraints for clean energy technologies.[7]

The current global recycling rate for end-of-life LIBs remains comparatively low, with industry estimating within 5% are properly processed, while the majority of valuable materials end up in landfills annually.[8] To address the challenges mentioned above, it is imperative to attribute importance to the spent LIBs recycling industry. This article discusses the primary industrial methods of spent LIBs recycling, introducing their advantages and limitations, while stimulating broader societal involvement in the LIB recovery ecosystem.

2 Spent LIBs recycling procedure

The recycling process for LIBs typically proceeds through several stages. Initially, batteries undergo discharge to eliminate residual energy and mitigate safety hazards. This step involves immersing the batteries in a saturated saline solution (such as NaCl) to completely discharge chemical energy.[9,10] The following pretreatment process consists of several approaches, separating and gathering different constituents. The final metal extraction stage typically employs chemical processes to conduct the metallurgical procedure, recovering valuable metals, alloys, and metallic salts.[11] The dominant industrial approaches currently remain pyrometallurgy and hydrometallurgy. In recent years, an innovative process named mild recycling has been developed, which integrates the advantages of both conventional methods[7,12]

2.1 Pretreatment

Battery materials are fully encapsulated and composite, requiring thorough separation into purified material streams. This process ameliorates metal recovery rates while minimizing energy consumption in subsequent processing stages.[13]

Industry typically employs mechanism separation, which reduces components to smaller fragments through shredding processes. To ensure operational safety, this step is conducted under an inert atmosphere (such as N_2), with continuous water spraying for temperature management.[2] The following separation techniques generally include magnetic separation, density-based classification, and froth flotation, extracting metal-bearing particles.[13]

An alternative approach employs thermal pretreatment, which exhibits operational simplicity.[14] After the battery casing is removed, the residual components undergo pyrolysis at temperatures exceeding 600 ° C. During this process, binders and flammable organic components are controllably deactivated and decomposed. Subsequently, metal-enriched particles can be extracted from thermally treated materials through seizing processes.[6] Notably, the flue gases generated from organic combustion require an additional purification procedure to eliminate toxic emissions.[15]

2.2 Pyrometallurgy methods

Pyrometallurgy is the most mature and widely adopted method in industrial lithium battery recycling. Its operational procedure is relatively simple, making it suitable for large-scale production.[13,16] The feedstock can be either untreated spent LIBs or metal-bearing particles. During smelting processes exceeding 1400°C, the waste materials melt and are converted to alloys, while other impurities volatilized into gases or absorbed into the slag.[2,17] Figure 1[16] illustrates a pyrometallurgy facility.

Pyrometallurgy relies on high temperature decomposition and material extraction, resulting in significant energy consumption and high operational costs.[2] Additionally, only cobalt, nickel, copper, and iron can be effectively recovered, with lithium, aluminum and manganese partition into the slag, requiring additional hydrometallurgical process for recovery.[18] This incurs high operational costs and presents significant challenges in addressing lithium resource scarcity.[12] Furthermore, the smelting process generates substantial toxic and greenhouse gas emissions, necessitating additional gas purification systems.[13]

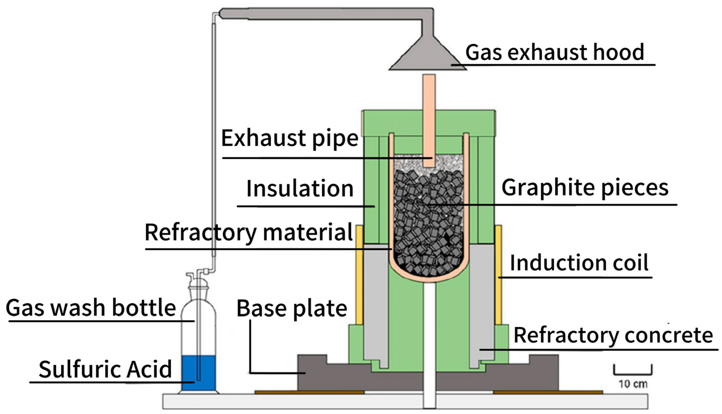

Figure 1: InduMelt reactor

2.3 Hydrometallurgy methods

Hydrometallurgy primarily consists of leaching and separation processes, where leaching is considered the most crucial stage. Various leaching methodologies have been developed, including inorganic acid leaching, organic acid leaching, ammonia leaching, and alkaline leaching.[12,13] Industrially, inorganic acids such as sulfuric acid are predominantly employed due to their inexpensive cost. However, they present challenges in the recovery of waste acid and can potentially contribute to secondary pollution.[19] While organic acids demonstrate comparable leaching efficiency with enhanced environmental benefits, their implementation is restricted by higher costs and reduced economic feasibility.[20] Figure 2[13] presents a schematic overview of the hydrometallurgical process.

A fundamental limitation of acid leaching lies in its lack of metal selectivity, which increases the complexity of the separation and purification processes.[20] In contrast, alkaline leaching demonstrates a particular ability in the extraction of aluminum, while ammonia-based leaching shows superior selectivity for nickel, cobalt, and copper.[21,22]

Solvent extraction is the most commonly used method for separating metals from leaching solutions, based on the different solubility of metals in aqueous and organic phases.[23,24] Meanwhile, due to the complexity of the leaching solution, two or more extraction agents are generally employed to enhance metal selectivity and extraction efficiency.[4]

Hydrometallurgy processing is typically conducted at moderate temperatures (200°C) with relatively short reaction time, leading to lower energy consumption compared to pyrometallurgy methods.[2] However, the process requires pretreated particles as feedstock, and involves complex operations. The separation of mixed metal ions from solu-

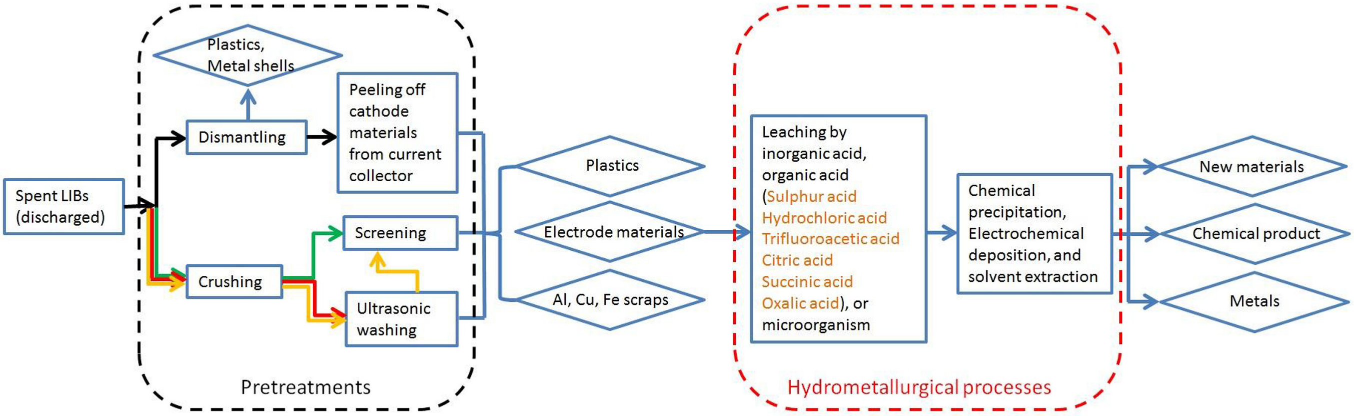

Figure 2: Flowchart of typical hydrometallurgy

tion presents significant technical challenges, particularly in the recovery of lithium, which is highly dependent on the optimization at each stage of the process.[25] Furthermore, the substantial quantities of chemical reagents employed in this process may result in secondary environmental contamination, simultaneously posing a risk of equipment corrosion.[23] It is crucial to implement reagent recovery and degradation processes, to mitigate environmental footprint and enhance economic viability.[26]

2.4 Mild recycling methods

In recent years, a mild recycling approach has emerged. This innovative technique combines the advantages of both conventional processes. Industrial applications primarily involve vacuum carbothermal reduction and Sulfation roasting.[27] The former process begins with roasting of pretreated materials and carbon at 700°C under reduced pressure conditions, where carbothermic reduction converts the feed into metal oxides (Co/Ni/Mn) and lithium carbonate.[12] Subsequently, hydrometallurgy methods are implemented for separating individual metal components, while lithium carbonate recovery can be achieved through simple water leaching.[28] This carbothermal reduction process demonstrates high selectivity and efficiency in lithium extraction. Furthermore, the reduction in the oxidation states of the transition metals during roasting significantly decreases the dosage of reducing agents in the separation process.[7,29]

The latter method involves roasting materials with sulfation agents at approximately 400°C, converting valuable metals into soluble sulfates. Subsequent water leaching and solvent extraction enable the separation of various metals.[30] However, the separation of lithium from transition metals remains complex. The elevation of roasting temperatures can address the problem, inducing the decomposition of transition metal sulfates into insoluble oxides, thereby enhancing selectivity to lithium sulfate.[28]

The mild recycling approaches significantly reduces energy and reagent consumption compared to traditional methods, while achieving comprehensive recovery of all valuable metallic constituents.[12] This innovative method shows considerable promise for LIBs recycling, contributing to sustainable provision of critical material of LIBs, with minimal environmental footprint. It necessitates additional fundamental research and industrial validation to realize its full potential.

3 Conclusion

The challenges presented by spent lithium-ion batteries necessitate efficient recycling solutions that balance technical feasibility, economic viability, and environmental sustainability. While pyrometallurgy and hydrometallurgy approaches are industrially established, their inherent limitations constrain sustainable scalability. The emerging mild recycling methodology presents a promising integration of both conventional techniques, optimizing energy consumption while maximizing material recovery.

Scaling these methods necessitates collaboration across policy, industry, and academia to establish effective recycling infrastructure. Furthermore, supportive regulatory frameworks and economic incentives are essential to stimulate industrial investment and innovation. By integrating technological development with systemic societal engagement, a circular economy for LIBs can mitigate resource depletion, reduce environmental harm, and sustain the clean energy transition.

References

[1] Harper, G.; Sommerville, R.; Kendrick, E.; Driscoll, L.; Slater, P.; Stolkin, R.; Walton, A.; Christensen, P.; Heidrich, O.; Lambert, S.; others *Nature* **2019**, *575*, 75–86.

[2] Ma, X.; Meng, Z.; Bellonia, M. V.; others *Nat. Rev. Clean Technol.* **2025**, *1*, 75–94.

[3] Xu, C.; Dai, Q.; Gaines, L.; others *Commun Mater* **2020**, *1*, 99.

[4] Lombardo, G.; Ebin, B.; St. J. Foreman, M. R.; Steenari, B.-M.; Petranikova, M. *ACS Sustainable Chemistry & Engineering* **2019**, *7*, 13668–13679.

[5] Bae, H.; Kim, Y. *Materials Advances* **2021**, *2*, 3234–3250.

[6] Makuza, B.; Tian, Q.; Guo, X.; Chattopadhyay, K.; Yu, D. *Journal of Power Sources* **2021**, *491*, 229622.

[7] Yang, Y.; Okonkwo, E. G.; Huang, G.; Xu, S.; Sun, W.; He, Y. *Energy Storage Materials* **2021**, *36*, 186–212.

[8] Shi, J.; Peng, C.; Chen, M.; others *JOM* **2019**, *71*, 4473–4482.

[9] Zeng, X.; Li, J.; and, N. S. *Critical Reviews in Environmental Science and Technology* **2014**, *44*, 1129–1165.

[10] Nie, H.; Xu, L.; Song, D.; Song, J.; Shi, X.; Wang, X.; Zhang, L.; Yuan, Z. *Green Chem.* **2015**, *17*, 1276–1280.

[11] Zheng, X.; Zhu, Z.; Lin, X.; Zhang, Y.; He, Y.; Cao, H.; Sun, Z. *Engineering* **2018**, *4*, 361–370.

[12] Liu, C.; Lin, J.; Cao, H.; Zhang, Y.; Sun, Z. *Journal of Cleaner Production* **2019**, *228*, 801–813.

[13] Huang, B.; Pan, Z.; Su, X.; An, L. *Journal of Power Sources* **2018**, *399*, 274–286.

[14] Xu, J.; Thomas, H.; Francis, R. W.; Lum, K. R.; Wang, J.; Liang, B. *Journal of Power Sources* **2008**, *177*, 512–527.

[15] Zhang, G.; Du, Z.; He, Y.; Wang, H.; Xie, W.; Zhang, T. *Sustainability* **2019**, *11*, 2363.

[16] Wang, J.; Ma, J.; Zhuang, Z.; Liang, Z.; Jia, K.; Ji, G.; Zhou, G.; Cheng, H.-M. *Chemical Reviews* **2024**, *124*, 2839–2887, PMID: 38427022.

[17] Georgi-Maschler, T.; Friedrich, B.; Weyhe, R.; Heegn, H.; Rutz, M. *Journal of Power Sources* **2012**, *207*, 173–182.

[18] Sun, Z.; Cao, H.; Xiao, Y.; Sietsma, J.; Jin, W.; Agterhuis, H.; Yang, Y. *ACS Sustainable Chemistry & Engineering* **2017**, *5*, 21–40.

[19] Li, Y.; Lv, W.; Huang, H.; Yan, W.; Li, X.; Ning, P.; Cao, H.; Sun, Z. *Green Chemistry* **2021**, *23*, 6139–6171.

[20] Asadi Dalini, E.; Karimi, G.; Zandevakili, S.; Goodarzi, M. *Mineral Processing and Extractive Metallurgy Review* **2021**, *42*, 451–472.

[21] Ku, H.; Jung, Y.; Jo, M.; Park, S.; Kim, S.; Yang, D.; Rhee, K.; An, E.-M.; Sohn, J.; Kwon, K. *Journal of Hazardous Materials* **2016**, *313*, 138–146.

[22] Sun, Z.; Xiao, Y.; Sietsma, J.; Agterhuis, H.; Yang, Y. *Environmental Science & Technology* **2015**, *49*, 7981–7988.

[23] Fan, E.; Li, L.; Wang, Z.; Lin, J.; Huang, Y.; Yao, Y.; Chen, R.; Wu, F. *Chemical Reviews* **2020**, *120*, 7020–7063, PMID: 31990183.

[24] Virolainen, S.; Fini, M. F.; Laitinen, A.; Sainio, T. *Separation and Purification Technology* **2017**, *179*, 274–282.

[25] Valio, J. Johanna Valio CRITICAL REVIEW ON LITHIUM ION BATTERY RECYCLING TECHNOLOGIES Master ´s Programme in Chemical, Biochemical and Materials Engineering Major in Functional Materials Master's thesis for the degree of Master of Science in Technology. 2017.

[26] Harper, G. D.; Kendrick, E.; Anderson, P. A.; Mrozik, W.; Christensen, P.; Lambert, S.; Greenwood, D.; Das, P. K.; Ahmeid, M.; Milojevic, Z.; others *Journal of Physics: Energy* **2023**, *5*, 021501.

[27] Yue, Y.; Wei, S.; Yongjie, B.; Chenyang, Z.; Shaole, S.; Yuehua, H. *ACS Sustainable Chemistry & Engineering* **2018**, *6*, 10445–10453.

[28] He, M.; Jin, X.; Zhang, X.; Duan, X.; Zhang, P.; Teng, L.; Liu, Q.; Liu, W. *Green Chem.* **2023**, *25*, 6561–6580.

[29] Hu, J.; Zhang, J.; Li, H.; Chen, Y.; Wang, C. *Journal of Power Sources* **2017**, *351*, 192–199.

[30] Zhong, Y.; Li, Z.; Zou, J.; Pan, T.; Li, P.; Yu, G.; Wang, X.; Wang, S.; Zhang, J. *Journal of Hazardous Materials* **2024**, *474*, 134794.